120 Days of Quantitative Finance

Month 4: Stochastic Calculus

Week 14: SDEs and Black-Scholes

Day 97: Merton Jump-Diffusion Model

1 Introduction

The Black-Scholes model assumes continuous price evolution. However, real markets exhibit sudden jumps due to unexpected events.

The Merton Jump-Diffusion model extends GBM by incorporating jumps.

2 Model Definition

The price dynamics are given by:

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_t(J-1)dN_t$$

where:

- W_t is Brownian Motion
- N_t is a Poisson process with intensity λ
- J is the random jump size

3 Interpretation

- Diffusion term models normal market fluctuations

- Jump term captures rare but significant shocks

4 Discrete Simulation

The process can be simulated as:

$$S_{t+\Delta t} = S_t \cdot \exp\left((\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\epsilon_t\right) \cdot J^{N_t}$$

5 Jump Process

$$N_t \sim \text{Poisson}(\lambda \Delta t)$$

$$J = \exp(\mu_J + \sigma_J Z)$$

where $Z \sim \mathcal{N}(0, 1)$.

6 Key Features

- Discontinuous price paths
- Fat-tailed return distributions
- Better modeling of extreme events

7 Financial Importance

Jump-diffusion models are used to:

- capture crash risk
- improve option pricing accuracy
- model real market behavior

8 Conclusion

The Merton model provides a more realistic alternative to GBM by incorporating jumps, making it highly relevant for practical financial modeling.